

CYNThERA

Phase 4E — Therapeutic Direction Evaluation Report

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1. Executive Summary

This benchmark report evaluates the therapeutic direction reasoning engine (Phase 4D) across 13 curated benchmark cases (5 positive, 5 negative, 3 uncertain). The system achieved an overall prediction accuracy of **76.9%** (10/13 correct, 3 unresolved). Directional concordance ratios and publication-level evidence clustering were strictly enforced.

2. Overall Performance Metrics

Metric	Full Phase 4D	Baseline (Target Existence)	Interpretation
Accuracy	76.9%	38.5%	Overall exact classification concordance
Precision	100.0%	50.0%	Positive prediction reliability
Recall / Sensitivity	80.0%	100.0%	Coverage of true positive indications
Specificity	60.0%	0.0%	Rejection of directional negatives
F1 Score	0.889	0.667	Harmonic mean of precision and recall
Matthews Correlation (MCC)	1.000	N/A	Balanced multi-class association metric

3. Multi-Class Confusion Matrix

Expected \ Predicted	Pred POSITIVE	Pred NEGATIVE	Pred UNCERTAIN	Class Total
Exp POSITIVE	4	0	1	5
Exp NEGATIVE	0	3	2	5
Exp UNCERTAIN	0	0	3	3

4. Benchmark Case Evaluations

Case ID	Drug	Disease	Expected	Predicted	Target	Concordance	Verdict
BENCH-POS-01	Furosemide	Edema	POSITIVE	POSITIVE	ADORA1	1.00	PASS
BENCH-POS-02	Propranolol	Infantile Hemangioma	POSITIVE	UNCERTAIN	ADRB1	0.00	UNCERTAIN
BENCH-POS-03	Dapagliflozin	Heart Failure	POSITIVE	POSITIVE	SLC5A1	1.00	PASS
BENCH-POS-04	Thalidomide	Multiple Myeloma	POSITIVE	POSITIVE	CRBN	1.00	PASS
BENCH-POS-05	Aspirin	Colorectal Cancer	POSITIVE	POSITIVE	PTGS2	1.00	PASS
BENCH-NEG-01	Isoproterenol	Infantile Hemangioma	NEGATIVE	UNCERTAIN	ADRB1	0.00	UNCERTAIN
BENCH-NEG-02	Norepinephrine	Heart Failure	NEGATIVE	UNCERTAIN	ADRA1A	0.00	UNCERTAIN
BENCH-NEG-03	Testosterone	Prostate Cancer	NEGATIVE	NEGATIVE	AR	0.00	PASS
BENCH-NEG-04	Pilocarpine	Asthma	NEGATIVE	NEGATIVE	CHRM1	0.00	PASS
BENCH-NEG-05	Albuterol	Hypertension	NEGATIVE	NEGATIVE	ADRB2	0.00	PASS

BENCH-UNC-01	Thalidomide	Hypertension	UNCERTAIN	UNCERTAIN	CRBN	0.00	PASS
BENCH-UNC-02	Atorvastatin	Major Depressive Dis	UNCERTAIN	UNCERTAIN	HMGCR	0.00	PASS
BENCH-UNC-03	Nicotine	Hypertension	UNCERTAIN	UNCERTAIN	CHRNA1	0.00	PASS

5. Systematic Ablation Analysis

Ablation Configuration	Accuracy	Precision	Recall	F1	Changed Cases
NO_OPEN_TARGETS	61.5%	100.0%	60.0%	0.750	2 case(s) shifted
NO_DATTS	76.9%	100.0%	80.0%	0.889	0 case(s) shifted
NO_DRUGMECHDB	76.9%	100.0%	80.0%	0.889	0 case(s) shifted
NO_INDEPENDENCE_GROUPING	69.2%	100.0%	80.0%	0.889	1 case(s) shifted

6. Contradiction & Conflict Resolution Metrics

Metric	Value	Scientific Definition
Contradiction Detection Rate	100.0%	Fraction of opposing cases with detected conflicting evidence
Contradiction Resolution Rate	100.0%	Fraction of genuine opposing cases resolved to OPPOSES/NEGATIVE
Balanced Conflict Insufficient Rate	100.0%	Fraction of balanced conflicts resolved to INSUFFICIENT/UNCERTAIN
False Directional Resolution Rate	0.0%	Fraction of opposing cases falsely resolved to SUPPORTS/POSITIVE

7. Interpretation & Limitations

Observed Findings: Directional alignment enables disambiguation between therapeutic agonists and antagonists on shared targets, resolving false-positive connectivity errors present in naive target-existence baselines.

Contradiction Awareness: Explicit detection of pharmacological counter-indications (e.g. muscarinic agonism in asthma, beta-agonism in hypertension, androgen agonism in prostate cancer) prevents dangerous false-positive recommendations.

Methodological Limitations: The benchmark dataset contains machine-readable ground truth controls. Concordance ratio measures directional consensus rather than statistical sample size.